OBJECT ORIENTED SOFTWARE ENGINEERING

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Architectural Styles…

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Architectural Styles

1. **Data-centered architectures**
2. **Data flow architectures**
3. **Call and return architectures**
4. **Object-oriented architectures**
5. **Layered architectures**

3. Call and Return Architecture Style

This architectural style enables to achieve a program structure that is relatively easy to modify and scale.

- A number of subsyles exist within this category:

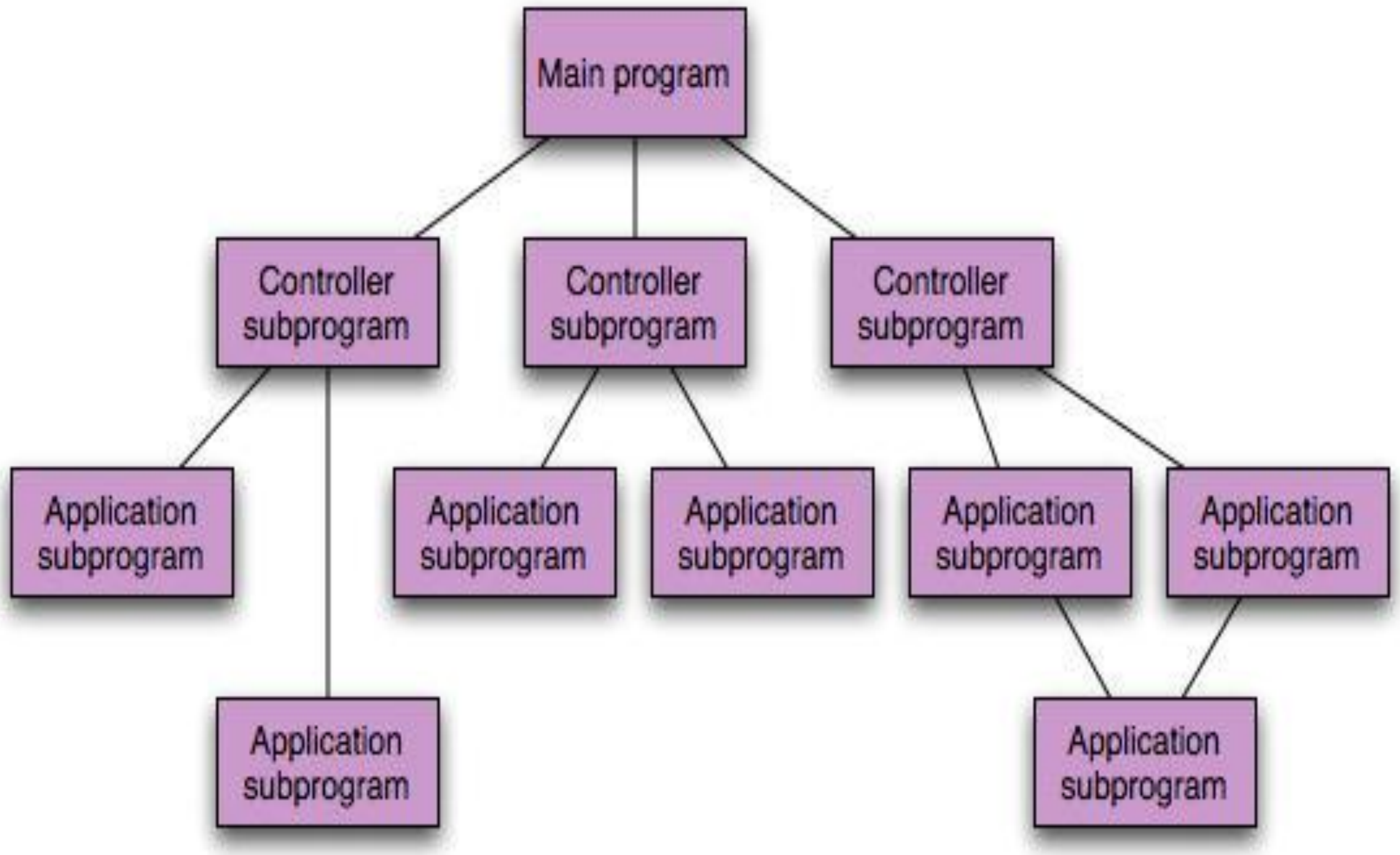
1. Main program / subprogram architectures –

- This classic program structure **decomposes function into a control hierarchy** where a **"main"** program **invokes** a number of program components that in-turn may invoke still other components.

2. Remote procedure call architectures –

- The components of a main program / subprogram architecture are distributed across multiple computers on a network

3. Call and Return Architecture Style…



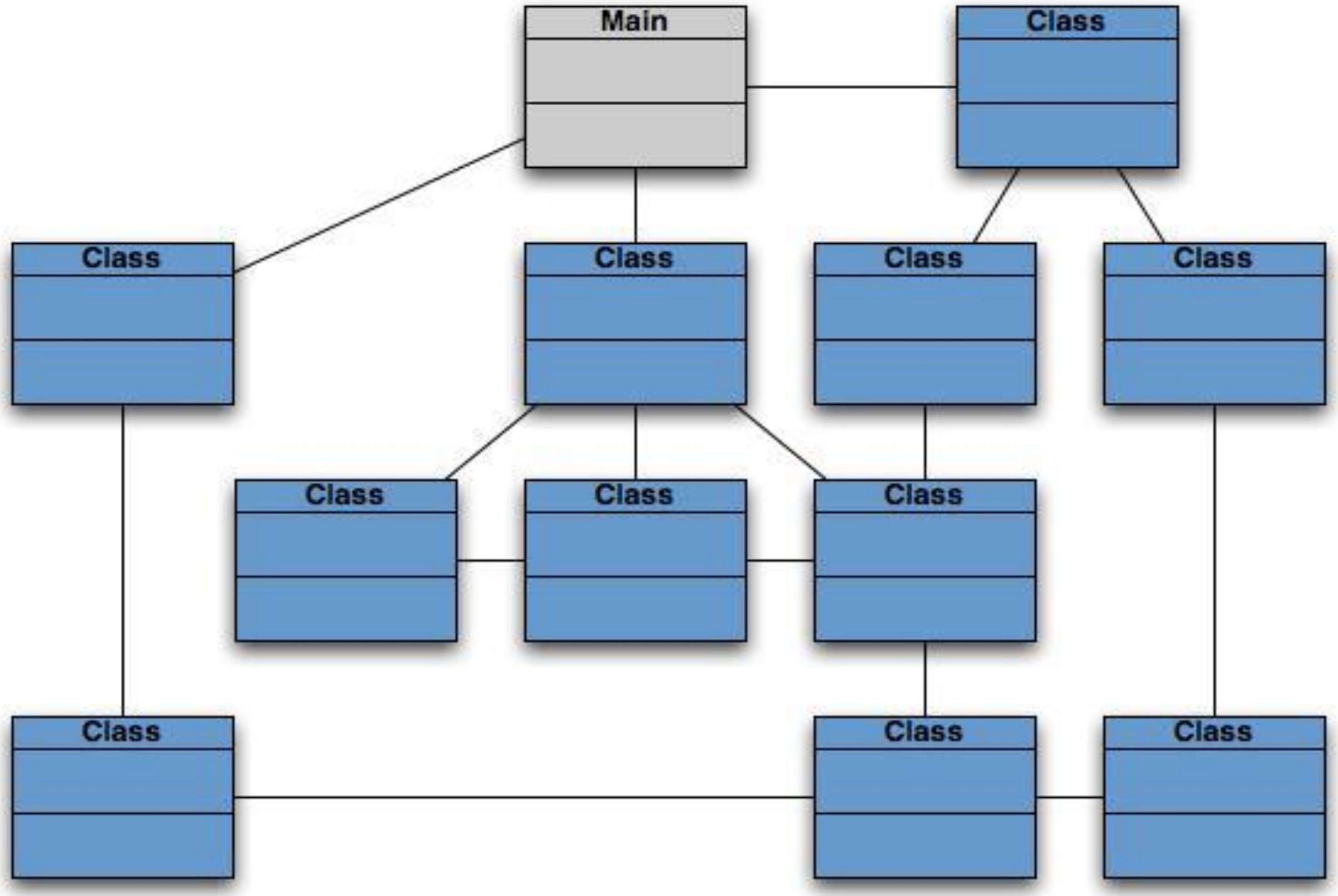
4. Object Oriented Architecture Style

- Object-Oriented architecture maps the application to real world objects for making it more understandable.

- It is easy to **maintain and it improves** the quality of the system due to **code reusability.**

- This architecture provides **reusability through polymorphism** and **abstraction**.

4. Object Oriented Architecture Style…



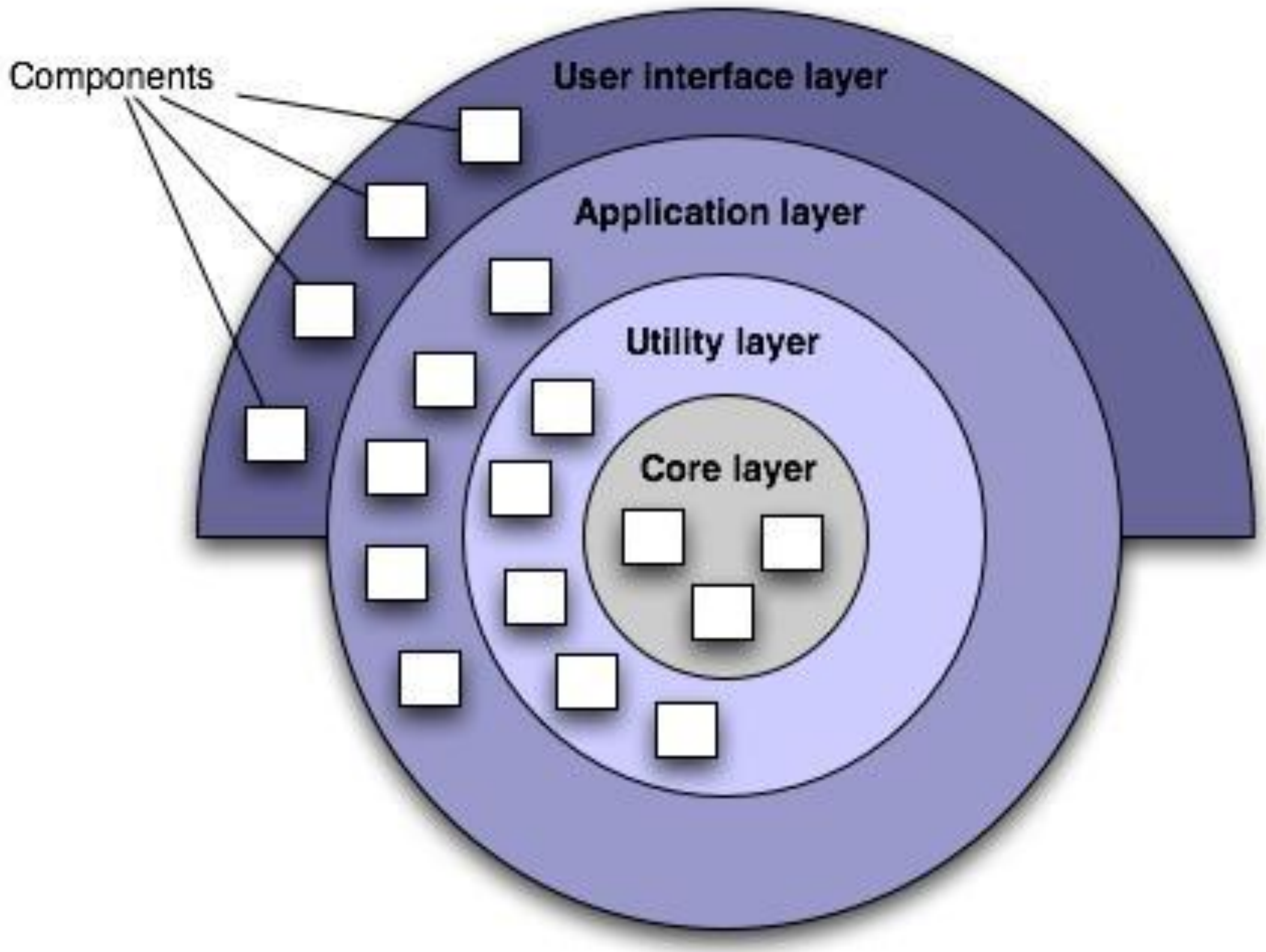
4. Object Oriented Architecture Style…

- It has ability to **manage the errors** during execution. (Robustness)
- It has ability to **extend new functionality** and **does not affected** on the system.
- It improves **testability** through **encapsulation.**
- Object-Oriented architecture **reduces** the **development time and cost**.

5. Layered Architecture Style

- A number of different layers are defined, each accomplishing operations that progressively become closer to the machine instruction set.
  - At the outer layer, **components service user interface operations.**
  - At the inner layer, **components perform operating system interfacing.**
  - Intermediate layers provide **utility services and application software functions**.

5. Layered Architecture Style



5. Layered Architecture Style…

- These architectural styles are only a small subset of those available.

- Once requirements engineering uncovers the characteristics, and constraints can be chosen. In many cases, **more than one pattern might be appropriate and alternative architectural styles can be designed and evaluated**.
  - **For example**: a layered style can be combined with a data-centered architecture in many database applications.

THANK YOU

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